

# Salavat Galimov

Raleigh, NC | (910) 939-0267 | [salgalia@proton.me](mailto:salgalia@proton.me) | [trade42.github.io](https://github.com/trade42) | US citizen

**Seeking:** internship from December 2026 through Summer 2027; open to relocating to San Francisco

**Flight videos and brace CAD:** <https://drive.google.com/drive/folders/158aF8HUKg1S8hqNXXOVQKNSS-r63nkcE>

*EE student who builds electromechanical systems end to end: CAD, fabrication, wiring, firmware, bring-up, and field testing.*

## EDUCATION

**NC State University, Raleigh, NC** | BS Electrical Engineering, second-year (junior standing by credit hours) **Expected May 2029**

Coursework: mechanics; electricity and magnetism; signals, circuits and systems; computer systems; multivariable calculus.

## MECHATRONICS & HARDWARE EXPERIENCE

**Drop Foot Active Brace** | Wearable Robotic Assistive Device (solo design, build, and bring-up) **2026 – Present**

- Delivered reliable active foot lift for an adult client with drop foot by building a custom robotic brace: 3 iterations from concept to on-body testing in under 30 days.
- Engineered the paracord cable drive that turns servo rotation into foot lift, plus the 3D-printed ankle saddle and motor mounts (full CAD assembly); fabricated, assembled, and wired each prototype.
- Integrated dual 40 kg-cm servos, a thigh-mounted FSR gait sensor, and 3S Li-ion power on an Arduino Nano; wrote closed-loop C/C++ firmware that calibrates the FSR input and commands the servos.
- Ran 6+ debug iterations across power, sensor, and actuator subsystems, including a 5 V rail brownout fix; tested a 12 V DC motor + L298N H-bridge as an alternative actuator.

**Robotics Design Challenge: Lab Automation Cell** | Anthrogen (YC S24), Final Interview Round **2026**

- Designed a full lab automation cell in Fusion 360, evolved from an earlier X-rail pipetting gantry: instrument stations, per-station robotic handling, and automated sample transport between stations.

**Aerobatic RC Plane** | From-Scratch Design, Build, and Flight Test **2021 – 2023**

- Placed 4th at Aerospace Russian Nationals (Camsat Russia, 130+ teams) with a 1.75 m aerobatic RC plane built from scratch (foam airframe, carbon-fiber spars, brushless motors, LiPo); 4 design cycles, 20+ test flights.

**Two-Stage Solid-Fuel Rocket with Onboard Telemetry** | Aerospace Club **2021 – 2023**

- Placed 2nd at Aerospace Russian Nationals (Camsat Russia, 130+ teams) with a two-stage rocket that reached ~5,000 m apogee; designed in SolidWorks, built, and flown with an onboard IMU, altitude telemetry, and stabilization sensors.

**Stealth Startup** | AI Topology Optimization for UAV Airframes (Founder) **2026 – Present**

- Building an AI platform that turns rough sketches into weight-optimized UAV airframes and exports them straight to SolidWorks; a GNN surrogate for finite element analysis scores stress on 1,000+ candidates.

## LEADERSHIP & COMPETITIONS

**Technology Student Association (TSA)** | UAV/Drone Team Lead; VEX Robotics Captain **2023 – 2025**

- Led the Drone/UAV team to 1st at state and 7th nationally: airframe design, build, flight tuning, and mission prep. Captained the chapter's VEX Robotics team to 1st at state.

**Aerospace Club** | Flight Projects Lead **2021 – 2023**

- Led the UAV team to 3rd at Aerospace Russian Nationals (Camsat Russia, 130+ teams), flying autonomous missions.

## SOFTWARE PROJECTS

**Vale** | Stealth Prop Trading Firm (Founder) **2026 – Present**

- Built a Rust research engine with 8,800+ tests behind a 5-lane CI gate; strategies must clear a statistical validation gate that rejects results explainable by chance, then a sealed out-of-sample holdout.
- Live tick-data pipeline across 25 crypto instruments; every research trial is logged to a hash-chained audit ledger.

**Vext** | Real-Time Multi-AI Platform (solo build) **2026**

- Async Rust (Axum, Tokio) backend streaming tokens to a Next.js frontend over WebSockets; 47+ Rust tests.

## TECHNICAL SKILLS & AWARDS

**Mechanical & Fabrication:** CAD in SolidWorks, Fusion 360, OnShape, Blender; mechanism and mount design; cable-drive actuation; 3D printing; foam and carbon-fiber airframe fabrication; assembly.

**Electromechanical:** servos, brushless and DC motors, L298N H-bridge, FSR and IMU sensors, Li-ion and LiPo power, wiring and soldering, bring-up, bench and field testing.

**Embedded & Software:** Arduino (C/C++), Raspberry Pi (Linux), Rust, Python, TypeScript / React / Next.js, Git, Linux (Arch); AI-assisted engineering with Claude Code, including Fusion 360 through an MCP bridge.

**Awards:** Game Changers Hackathon 1st place (\$500, team captain); TSA state champion (Drone/UAV and VEX), 7th nationally.